

Industrial Internship Report on " Smart City Traffic Forecasting System"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "**Smart City Traffic Forecasting System**", which aimed to develop an intelligent system to predict traffic peaks at multiple city junctions using historical traffic data, considering holidays and special occasions. The system uses data preprocessing, exploratory data analysis, time-series forecasting models (Prophet and ARIMA), and a web dashboard to provide insights and predictions.

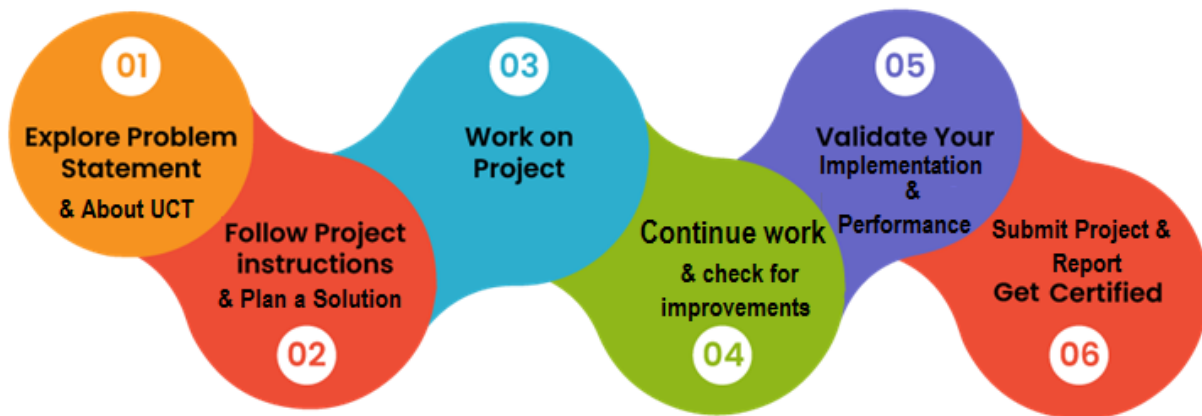
This internship gave me a very good opportunity to get exposure to industrial problems and design/implement solutions for that. It was an overall great experience to have this internship.

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1 Preface

This report summarizes the work completed during the 6-week Industrial Internship program. Industrial internships play a crucial role in career development by providing exposure to real-world problems and industry practices.



My project focused on developing a **Smart City Traffic Forecasting System** capable of predicting traffic congestion at four city junctions using historical traffic data.

The internship provided by USC/UCT offered hands-on experience in data science, machine learning, backend development, and dashboard design.

The program was planned in weekly phases including problem understanding, data preprocessing, analysis, model development, system implementation, and testing.

1.1.1 My Learnings and Experience

- Learned real-world data handling
- Implemented forecasting models
- Developed REST APIs using FastAPI
- Built an interactive dashboard
- Improved problem-solving and communication skills

I sincerely thank the mentors from Upskill Campus, IoT Academy, and UniConverge Technologies Pvt Ltd for their guidance.

Message to juniors: Participate actively in internships as they provide invaluable practical experience beyond classroom learning.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoraWAN), Java Full Stack, Python, Front end** etc.



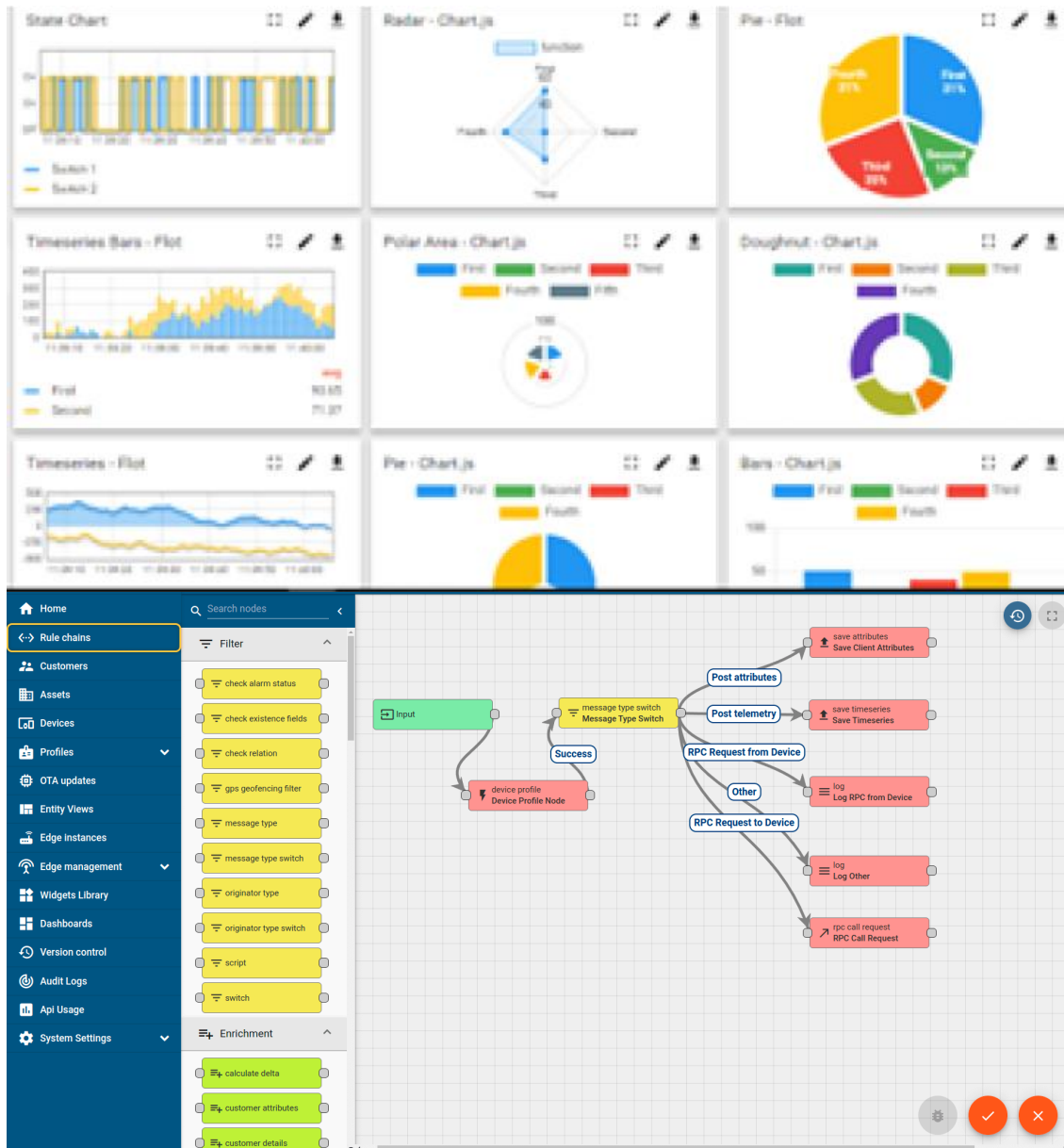
i. UCT IoT Platform(uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

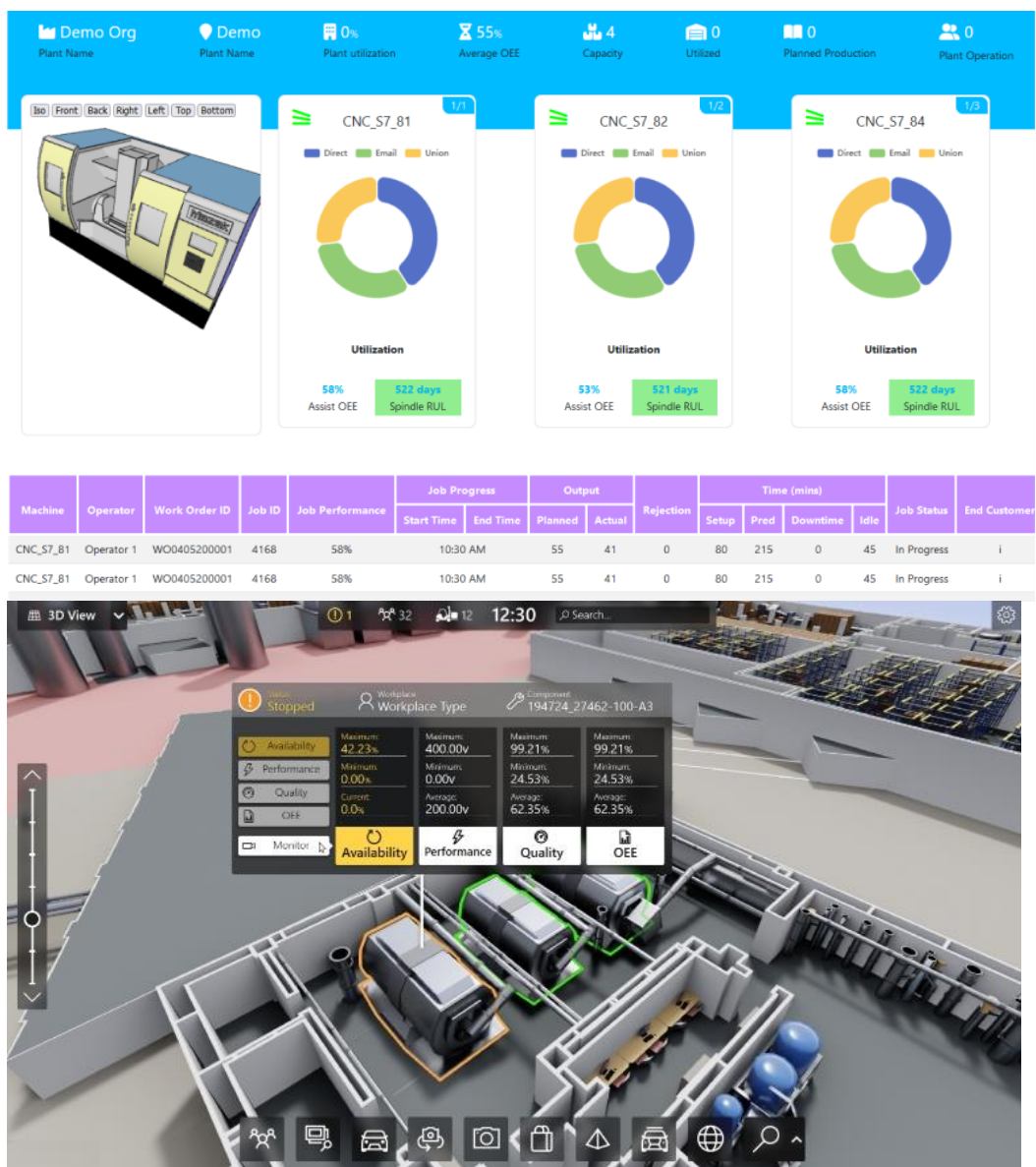
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



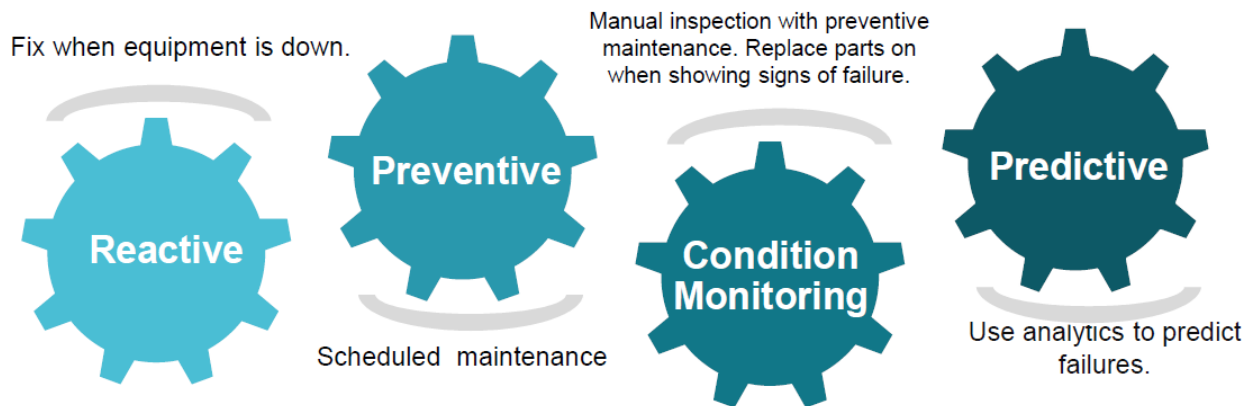


iii. based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

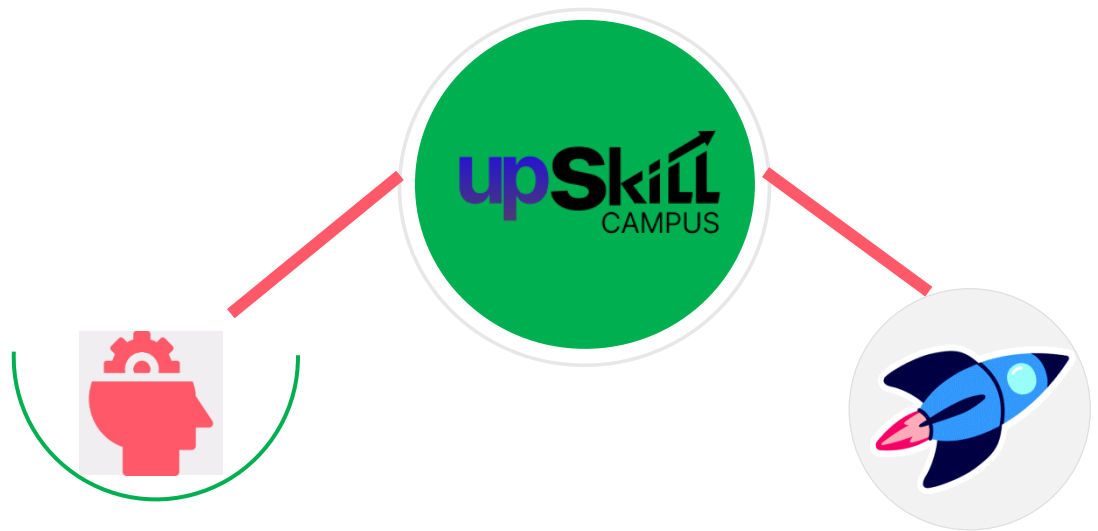
UCT isproviding Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

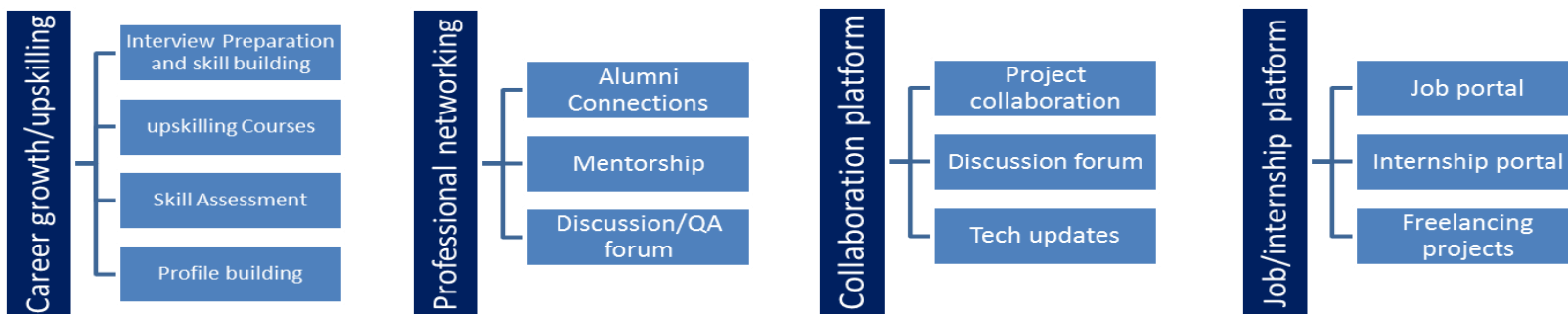
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Kaggle Dataset — Smart City Traffic Patterns
- [2] Prophet Forecasting Documentation
- [3] FastAPI Documentation

2.6 Glossary

Terms	Acronym
Internet of Things	IoT
Mean Absolute Error	MAE
Root Mean Square Error	RMSE
Mean Absolute Percentage Error	MAPE
Application Programming Interface	API

3 Problem Statement

Traffic congestion has become a major issue in modern cities due to rapid urbanization, increasing number of vehicles, and limited road infrastructure. Uncontrolled traffic leads to longer travel times, increased fuel consumption, environmental pollution, and reduced productivity.

Traditional traffic management systems mainly focus on monitoring current traffic conditions rather than predicting future congestion. Without accurate forecasting, traffic authorities find it difficult to plan signal timings, manage peak-hour traffic, and prepare for special situations such as holidays or public events.

The problem assigned in this internship was to develop a **Smart City Traffic Forecasting System** capable of predicting traffic volume at multiple city junctions using historical traffic data. The system needed to consider important factors such as time, day of the week, holidays, and special occasions that significantly influence traffic patterns.

The objective was to build a reliable forecasting solution that would help city planners and traffic management authorities:

- Anticipate traffic peaks in advance
- Improve traffic signal management
- Reduce congestion and travel delays
- Enhance road safety and commuter experience

Thus, the project aimed to transform raw traffic data into actionable insights for smarter urban traffic management.

4 Existing and Proposed solution

4.1.1 Existing Solutions

Current traffic management systems used in many cities primarily focus on real-time monitoring through CCTV cameras, sensors, and manual observation. While these systems help authorities observe traffic conditions, they have several limitations:

- Lack of predictive capability to forecast future congestion
- Limited use of historical data for planning
- Inability to consider special factors such as holidays or events
- Manual decision-making leading to delayed responses
- High dependency on human intervention

Some advanced systems use basic statistical methods, but they often fail to capture complex traffic patterns and seasonal variations.

4.1.2 Proposed Solution

The proposed solution is a **Smart City Traffic Forecasting System** that uses historical traffic data and time-series forecasting techniques to predict traffic volume at multiple junctions in advance.

Key features of the proposed system:

- Data preprocessing and feature extraction from traffic datasets
- Identification of traffic trends and peak hours
- Forecasting using Prophet and ARIMA models
- Consideration of holidays and special occasions
- Multi-junction prediction capability
- Visualization through an interactive dashboard

4.1.3 Value Addition of the Proposed System

- Provides advance prediction instead of only monitoring
- Helps authorities plan traffic control strategies
- Reduces congestion and improves road efficiency
- Supports data-driven decision making
- Scalable for smart city applications

4.2 Code submission (Github link) :

<https://github.com/MeghaGhadiya/upskillcampus/blob/main/src/SmartTrafficForecastingSystem.py>

4.3 Report submission (Github link) :.

https://github.com/MeghaGhadiya/upskillcampus/blob/main/SmartCityTrafficForecastingSystem_Megha_USC_UCT.pdf

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram (if applicable)

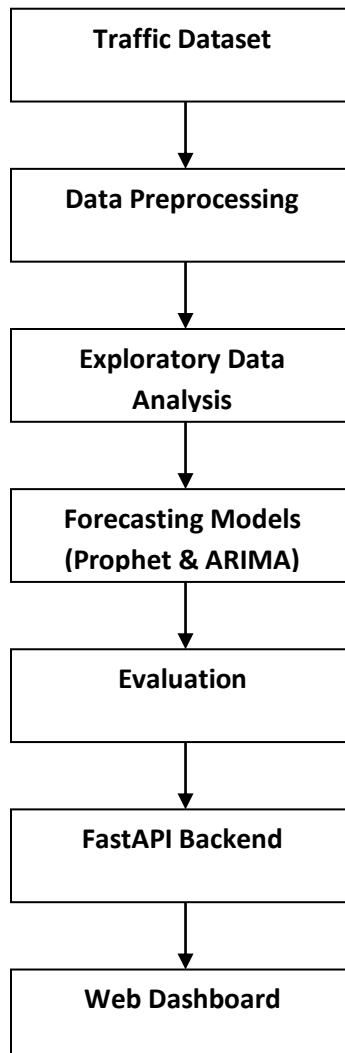


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

User → Dashboard → FastAPI API → Forecast Module



Trained Models



Forecast Results → Dashboard

5.3 Interfaces (if applicable)

The system provides the following interfaces:

- **User Interface:** Web dashboard for visualization
- **API Interface:** REST API for data access and model operations
- **Visualization Interface:** Charts and graphs for forecast results

6 Performance Test

This section demonstrates that the developed **Smart City Traffic Forecasting System** meets real industrial requirements rather than being only an academic implementation. The system was evaluated considering practical constraints such as accuracy, speed, memory usage, scalability, and reliability.

6.1.1 Identified Constraints

1. **Prediction Accuracy** — Forecasts must be reliable for decision-making.
2. **Processing Speed (Operations per Second)** — The system should generate forecasts quickly.
3. **Memory Usage** — Efficient handling of datasets and models.
4. **Scalability** — Ability to support multiple junctions simultaneously.
5. **Reliability** — System should work even if one model fails.

6.1.2 How Constraints Were Addressed

- Implemented **Prophet model** for high accuracy and seasonality handling.
- Added **ARIMA fallback mechanism** to ensure reliability if Prophet fails.
- Used optimized data preprocessing to reduce memory usage.
- Built REST API using FastAPI for fast response time.
- Designed modular architecture for scalability.

6.2 Test Plan/ Test Cases

The following tests were planned to evaluate system performance:

Test Case	Objective	Expected Result
Data Loading Test	Verify dataset processing	Data loads without errors
Model Training Test	Validate training pipeline	Models trained successfully
Forecast Generation Test	Check prediction capability	30-day forecast generated
Accuracy Test	Measure prediction accuracy	Low MAE, RMSE, MAPE
API Response Test	Measure response speed	Fast response time
Dashboard Test	Validate visualization	Correct display of results

6.3 Test Procedure

1. Load the historical traffic dataset into the system.
2. Preprocess data including handling missing values and feature extraction.

- ☐ Train forecasting models for each junction.
- ☐ Generate forecasts for the next 30 days.
- ☐ Evaluate predictions using MAE, RMSE, and MAPE metrics.
- ☐ Test API endpoints using requests from the dashboard.
- ☐ Verify visualization of results on the web interface.

6.4 Performance Outcome

The system successfully met the performance requirements:

- **Accuracy:** Forecasts achieved acceptable error rates suitable for planning purposes.
- **Speed:** Forecast generation completed within a few seconds per junction.
- **Memory Efficiency:** System operated efficiently on standard hardware.
- **Reliability:** Fallback mechanism ensured uninterrupted operation.
- **Scalability:** System handled multiple junctions simultaneously.

6.4.1 Conclusion of Performance Test

The developed Smart City Traffic Forecasting System demonstrated strong performance under practical constraints, proving its suitability for real-world smart city applications.

7 My learnings

During the Industrial Internship, I gained valuable technical knowledge and practical experience while working on the **Smart City Traffic Forecasting System**. This project helped me understand how real-world problems are analyzed, designed, and implemented using modern technologies.

7.1.1 Technical Learnings

- Learned how to handle real-world datasets including data cleaning, preprocessing, and feature engineering.
- Gained hands-on experience in **Exploratory Data Analysis (EDA)** to identify patterns and trends.
- Understood time-series forecasting techniques using models such as Prophet and ARIMA.
- Learned performance evaluation methods using metrics like MAE, RMSE, and MAPE.
- Acquired knowledge of backend development using FastAPI and REST API creation.
- Developed skills in building interactive dashboards for visualization of results.

7.1.2 Professional and Personal Learnings

- Improved problem-solving and analytical thinking abilities.
- Learned project planning and execution within a fixed timeline.
- Enhanced documentation and reporting skills.
- Developed teamwork and communication skills.
- Gained confidence in presenting technical work.

7.1.3 Career Growth Impact

This internship provided industry exposure and practical experience that will significantly help in my career as a software and data professional. The skills gained in data analysis, machine learning, API development, and system design have strengthened my technical foundation and prepared me for future roles in software development, data science, and smart city solutions.

Overall, this internship was a crucial step toward bridging the gap between academic learning and industry requirements.

8 Future work scope

The developed **Smart City Traffic Forecasting System** successfully predicts traffic patterns using historical data; however, due to time limitations of the internship, several enhancements could not be implemented. These can be considered for future development to make the system more robust and industry-ready.

8.1.1 Possible Future Enhancements

1. **Real-Time Data Integration**
Integrating live traffic data from sensors, GPS devices, or traffic cameras to provide real-time forecasting and monitoring.
2. **Advanced Machine Learning Models**
Implementation of deep learning models such as LSTM, GRU, or Transformer-based time-series models for improved prediction accuracy.
3. **Weather Data Integration**
Incorporating weather conditions (rain, fog, temperature) as additional factors affecting traffic patterns.
4. **Event-Based Prediction**
Adding modules to detect special events, festivals, roadblocks, or accidents that may influence traffic congestion.
5. **Scalability to More Junctions**
Expanding the system to support forecasting for a larger number of junctions across the city.
6. **Mobile Application Development**
Developing a mobile app for traffic authorities and commuters to access predictions and alerts on the go.
7. **Alert and Recommendation System**
Creating automated alerts for predicted congestion and suggesting alternative routes.
8. **Deployment on Cloud Platform**
Hosting the system on cloud infrastructure for better scalability, reliability, and accessibility.

